
Testing by betting when the bets are real: an e-value audit of 84 pre-registered market-efficiency hypotheses

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Abstract

E-values are motivated by a gambler betting against a null hypothesis. We report an audit where the gambler is not a metaphor: the null is a bookmaker’s posted price, the stake is a stake, and the wealth process *is* the e-process. The subject is a deployed UFC forecaster and a pre-registered search for pockets where it beats the closing line: 84 hypotheses over 1,787 priced bouts, originally analysed with paired log-loss and Benjamini–Hochberg. Re-asked as bets, three things change. (i) The 84 slices are overlapping subsets of one pool, so BH’s guarantee is not established; its one favourable discovery survives neither Benjamini–Yekutieli nor e-BH. (ii) A segment the fixed- n analysis reported as *failing to replicate* ($p = 0.30$) is, on the wealth scale, growing at the same rate in the replication window and 100 bouts short of $1/\alpha$: optional continuation converts a refutation into a schedule. (iii) A post-hoc threshold rule, paid for by a mixture martingale, survives the search it actually ran ($e = 31.3$) and dies twice over — once when the direction it could also have searched is priced in (15.6), and again when the bookmaker’s margin is charged (8.5). We report the ladder, not the winner.

1 A null hypothesis with money behind it

Testing by betting [Shafer, 2021, Grünwald et al., 2024, Vovk and Wang, 2021, Ramdas et al., 2023] introduces its central object through a gambler whose capital cannot grow in expectation if the null is true. The gambler is usually a device. Here it is the situation itself.

We audit VERTEX, a deployed UFC bout-outcome model, against a betting exchange’s closing line. The substrate is a walk-forward out-of-fold pool: 4,141 bouts scored between 2016 and 2026 by models refit at 32 quarterly origins, 1,787 of them across 225 events carrying an archived closing price. Two properties no simulation supplies:

The null is adversarial and external. H_0 is not an assumption of convenience — it is a number a counterparty posted with capital at risk, and a bettor who rejects it collects. Under H_0 the de-vigged closing probability q_t — the two closing decimals stripped of their overround by the power method: with raw implied probabilities $r = 1/d$, solve $r_a^k + r_b^k = 1$ and set $q = r^k$ — is the true conditional probability of the outcome $y_t \in \{0, 1\}$. **Predictability is enforced by the calendar.** The model probability p_t comes from a fit that saw only bouts strictly before its origin, and q_t is posted before the fight; neither has to be argued into being predictable.

The prior analysis of this pool [Prigodskii, 2026] declared 86 candidate segments, each with a stated mechanism for why a good market would be wrong in that exact place, in a registry¹ written before

¹Pre-registration here means author-controlled version history, not third-party registration: the registry is one inspectable source file whose commit (8f9fcbd) precedes the first scored artifact (6b5fbd9) in a public repository. That fixes the

any of them was scored. Two pairs are the same filter under different lenses, so the pre-registered family is 84; three further segments, added once results were visible, are marked post-hoc there and excluded from multiplicity accounting here — 84 + 3 is the 87 its title counts. It measured each with a composition-adjusted paired log-loss contrast, cluster-robust by event, and charged multiplicity with BH — an instrument whose own power section reports 10–15% against a 3 percentage-point edge, 0–1% *once BH multiplicity is charged*, and ~89,900 bouts needed. There are 1,787. That report is the reason for this one: the instrument was known to be inadequate before its answers were read.

2 The wealth process

For a bout with de-vigged closing probability q and outcome y , a predictable stake $\lambda \in (-1/(1 - q), 1/q)$ buys the increment $M = 1 + \lambda(y - q)$, and $\mathbb{E}_{H_0}[M] = 1$. The product over a segment's bouts *in calendar order* is a non-negative test martingale, so its terminal value is an e-value and, by Ville's inequality [Ville, 1939], $\Pr_{H_0}(\sup_t M_t \geq 1/\alpha) \leq \alpha$: it may be stopped at any time. Every segment filter is a function of pre-fight covariates, so membership is $\mathcal{F}_{t-1}$ -measurable, and restricting the product to a segment is optional skipping ($\lambda_t = 0$ off the segment), which preserves the martingale property. The growth-rate optimal stake under the alternative “the model's p is the truth” is available in closed form as $\lambda^* = (p - q)/[q(1 - q)]$, the model's edge deflated by the market's own variance. We report fractional Kelly at $\lambda = \frac{1}{2}\lambda^*$ and, because that fraction is a free choice, a predictable plug-in [Waudby-Smith and Ramdas, 2024] that learns the multiplier from past bouts only (§7).

The margin is a supermartingale, not an excuse. Write $f = \lambda q$ for the fraction of wealth the stake commits to the backed side and d for the decimal the book offers; a fair book would offer $d = 1/q$. Settling at d gives $\mathbb{E}_{H_0}[M] = 1 - f(1 - qd) \leq 1$, since $qd < 1$ whenever the book charges an overround: the deployable process is a valid e-value, uniformly conservative, short of the statistical one by exactly what the margin costs. On this pool the mean overround is 1.0482 and the shortfall 0.00525 nats per bout, so every e-value below is given at both prices, *fair* and *real* — a factor of five over 300 bouts at that rate, and seven on the segment below, whose stakes run larger.

Validation before use. Under a true null — prices and stakes held, only the outcome resampled from q — 40,000 replications give $\mathbb{E}[e] = 0.985 \pm 0.018$ per segment, $\Pr(\sup_t e_t \geq 1/\alpha)$ of 0.029/0.067/0.151 at $\alpha = 0.05/0.10/0.20$, none of the 84 above its own α , e-BH rejecting at all in 74 of the 40,000 runs, and the confidence sequence of §6 covering the truth at *every* t in 97.4% of them.

Where the model actually stands. Betting the model against the book over the whole priced pool gives $e = 3.6 \times 10^{-4}$ at fair odds and 3.1×10^{-8} at real ones. This is not a paper about beating a market, but about what can be honestly claimed inside a search that mostly failed.

3 The multiplicity charge BH was not licensed to make

The 84 slices are overlapping subsets of one pool of 1,787 bouts, scored by one model against one book: a single upset moves dozens of them together. BH's FDR guarantee requires independence or PRDS [Benjamini and Yekutieli, 2001], and neither is established here. Nor is the dependence of the benign kind that would make PRDS plausible without an argument: across the 3,486 segment pairs, membership correlations run from -0.49 to $+0.84$, half of them negative, and 83 pairs share no bouts at all. The classical repair is Benjamini–Yekutieli, which charges $\sum_{i \leq 84} 1/i = 5.01$. e-BH [Wang and Ramdas, 2022] is valid under *arbitrary* dependence and needs no dependence correction at all: its charge is K/α whatever the correlation structure turns out to be. On the discovery window the lab searched (1,191 bouts, the same 84 hypotheses):

procedure	guarantee holds?	rejects	
BH, paired statistic	not established	4	1 favourable, 3 against
Benjamini–Yekutieli ($\times 5.01$)	yes	0	
e-BH, wealth statistic	yes	0	cutoff $e \geq 1680$ ($k=1$)

hypotheses before they were scored, not before any contact with a pool already months in use — and no timestamp available to us would.

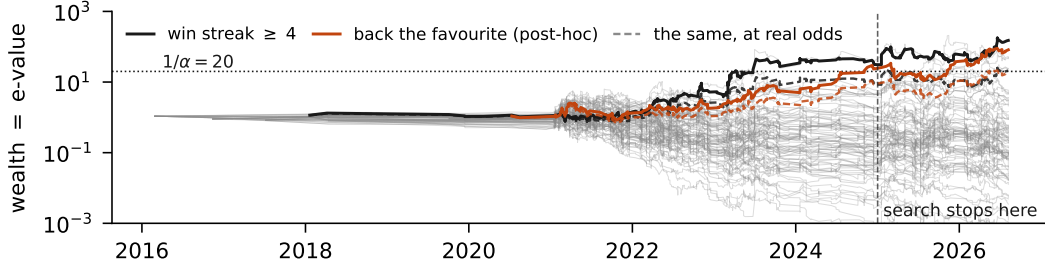


Figure 1: Wealth over calendar time for every pre-registered segment (grey) and the two the analysis singled out (coloured); dashed, the same bets at the book’s real prices. At the vertical rule the search stopped and the replication window began — a fixed- n analysis must start a new test to its right, a test martingale need not.

79 The lab’s single favourable discovery — a fighter carrying a win streak of four or more, BH $q =$
80 0.016 — does not survive either procedure whose guarantee actually holds. It would be wrong to
81 read this as e-values buying less. The same segment’s e-value on the full pool is 86.4 at fair odds
82 — the median over the pool’s five walk-forward seeds — and 12.3 at real ones, and for a *single*
83 pre-specified hypothesis $e \geq 20$ already rejects at $\alpha = 0.05$. What is expensive is not the machinery
84 but the charge for asking 84 questions of a pool this size — the original analysis’s own lesson, in a
85 currency where the arithmetic is visible.

86 4 “Failed to replicate” and “not yet” are different findings

87 The original analysis froze its discovery window, selected the streak segment, and ran a fresh fixed- n
88 test on 2025–26. It returned $p = 0.30$ and was reported, correctly under its own logic, as a failure
89 to replicate — the only move a fixed- n design has, since the second window buys a second test and
90 nothing carries over.

91 On the wealth scale the same 110 bouts read differently. Log-growth in the replication window is
92 $+0.0143$ nats per bout against $+0.0174$ in the window that selected it — 82% of the discovery rate,
93 the shrinkage one expects when a slice is chosen on the data that flatters it, but the same sign and
94 order. The e-value accumulated there is 4.81 — real evidence, short of $1/\alpha = 20$. At that rate,
95 crossing needs 210 bouts and 110 have arrived — an extrapolation at that same window’s rate, not
96 a guarantee: the e-process is valid whenever it is stopped, but the date it crosses on is a forecast.
97 The finding is not refuted; it is 100 bouts short, and at about 71 a year that is seventeen months
98 of continuing to multiply rather than choosing a stopping time in advance (Figure 1). At real odds
99 the window pays 2.48, the margin taking 0.0060 of the 0.0143, and the wait is another 253 bouts:
100 three and a half years. We report the pre-split wealth (31.4 fair, 8.5 real) but do not claim it: that
101 is the window in which the segment was chosen. The effect is stable across the pool’s five walk-
102 forward seeds — fair 56.2 to 150.9, real 8.4 to 21.2, medians 86.4 and 12.3 — a spread q -values
103 had no natural place to put. Only the most favourable seed clears $1/\alpha$ at real odds, and it is the one
104 decomposed above.

105 5 Pricing the freedom a post-hoc rule used

106 The largest effect in the original analysis is not a segment but a direction: where the model gives
107 the book’s favourite *more* than the book does, flat-staking that favourite returned $+11.5\%$ over 281
108 bets across 163 events — declared post-hoc there, because the cut at lean ≥ 0.05 was chosen after
109 seeing the sweep was positive from 0.02 to 0.22 and peaked at 0.13, and a fixed- n framework has
110 no principled penalty for that.

111 A mixture martingale has one, with weights that do not depend on which cut won [Robbins, 1970,
112 Howard et al., 2021]: each threshold’s wealth process is a martingale starting at 1, so any convex
113 combination is too, and the search is paid for by the prior mass not on the winner. The maximum
114 over thresholds, 119, is not a valid e-value. The uniform mixture over the 21 cuts is $e = 31.3$, which
115 clears $1/\alpha$. Widening the family to the 21 cuts the search could also have taken in the opposite

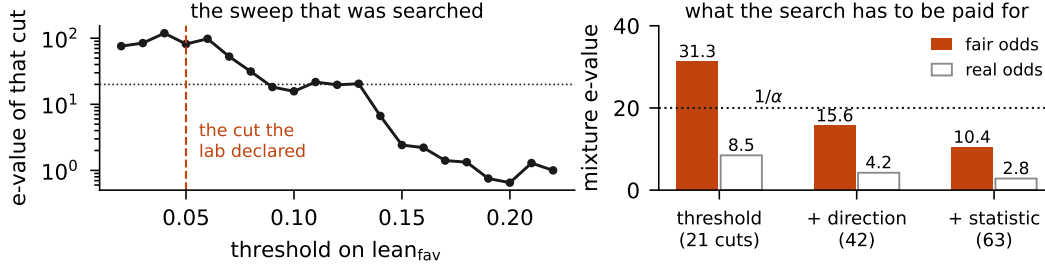


Figure 2: Left: the e-value of each threshold on the model’s lean toward the book’s favourite; the declared cut at 0.05 was chosen after this shape was visible. Right: a mixture over progressively more of the freedom the search had. At fair odds the finding pays for the threshold but not the direction; at real prices, neither.

116 direction gives 15.6, which does not. Adding the other disagreement statistic available on the same
 117 frame gives 10.4. Those are fair odds; at the prices the book offered, the same mixtures are 8.5, 4.2
 118 and 2.8 — a better paying the margin never clears $1/\alpha$ at all.

119 We regard the ladder, not any rung, as the result: it turns “this was post-hoc” from a caveat into a
 120 quantity. The finding absorbs the threshold search at fair odds, and neither the direction it could
 121 have taken nor the margin it would have paid.

122 6 What the sequential instrument needs

123 A growth rate makes planning a calculation rather than a verdict. Against a true 3 pp edge simulated
 124 on this pool’s own prices, the half-Kelly bet grows at 0.00166 nats per bout, so expected log-wealth
 125 reaches $\log(1/\alpha)$ in $\sim 1,800$ bouts — a 59% chance of having crossed by then, not a power level.
 126 Matched at 80% over 20,000 simulated runs, the sequential test needs 2,385 bouts to the paired
 127 log-loss test’s 5,346; at 5 pp, 864 to 1,902. It is roughly twice as cheap in fights, not because it is
 128 more powerful per observation, but because a paired contrast on a slice discards most of what each
 129 bout carries.

130 Anytime-validity is not free. On the directional rule the cluster bootstrap gives a 95% ROI interval
 131 of $[+2.2\%, +20.4\%]$, valid at the one n the analysis stopped at — and chosen after watching the
 132 number. The hedged betting confidence sequence [Waudby-Smith and Ramdas, 2024], valid at every
 133 n , gives $[-5.3\%, +22.9\%]$ and does not exclude zero.

134 7 Limitations

135 The stake and the de-vig are both free choices, and neither carries the result. Fractional Kelly at
 136 $0.25/0.5/1.0$ gives $e = 26.0/150.9/115.3$ on the streak segment’s primary seed and the predictable
 137 plug-in, which is no choice at all, settles at $c = 0.70$, $e = 63.1$; every variant clears $1/\alpha$. Under
 138 Shin’s de-vig the same three fair-odds figures — streak segment, replication window, ladder —
 139 read 120.6, 4.36, 34.0/17.0/11.3 against 150.9, 4.81, 31.3/15.6/10.4; the proportional split, which
 140 flatters a favourite-backing rule, would raise the ladder to 56.3/28.2/18.8, not lower it.

141 Boosting e-BH [Wang and Ramdas, 2022] would recover power but needs each e-value’s null dis-
 142 tribution, which a composite martingale null does not supply; pre-specified weights would be the
 143 better route, and were not.

144 Three contaminations are inherited from the pool and sized there: an age correction shipped on
 145 replication-window data; hyperparameters tuned on a window the pool later scores (122 of 1,191
 146 discovery bouts); a rating not fully point-in-time (21.9% of bouts, 0.0011 nats). Removing any
 147 moves the numbers unfavourably. And this is one sport, one book, one model: the construction
 148 generalises, the numbers do not.

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